

Research Paper Summary

Global Human Settlement Layer (GHSL) from Landsat Data

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Pesaresi et al. (2016), *European Commission Joint Research Centre*

1 Basic Information

- **Paper Title:** Operating Sentinel-2 and Landsat Sensors for Mapping Global Human Settlements
- **Authors:** Martino Pesaresi, Daniele Ehrlich, et al. (2016)
- **Dataset:** Global Human Settlement Layer (GHSL) produced by the European Commission Joint Research Centre (JRC).

2 What is this Paper About? (In Simple Words)

This paper describes the creation of the Global Human Settlement Layer (GHSL), a worldwide spatial dataset that maps built-up land coverage, buildings, and human structures at fine spatial resolutions (from 30 m up to 1 km).

- **Continuous Built-Up Surface:** Unlike vector maps that only show outlines, GHSL provides a continuous percentage grid showing the density of artificial structures and built-up area for every square kilometer of Earth.
- **Handling Rural/Remote Areas:** GHSL detects human-made structures even in remote rural corridors where road maps or volunteer databases might be incomplete.

3 How This Helps Our Project (PS26162)

3.1 1. Engineering the ‘Built-Up Density’ Feature

Vector maps like OSM may occasionally miss unmapped rural factories. GHSL provides a fallback continuous feature for our ML model:

Feature: Built_Up_Density = Percentage of built-up structural surface in a 1 km^2 cell (1)

- A thermal hotspot detected inside a high built-up density grid ($> 60\%$) has a significantly higher probability of being an **industrial plant**.
- A thermal hotspot detected in a 0% built-up grid is almost certainly a **wildfire or agricultural burn**.

3.2 2. Enhancing Spatial Fusion Beyond Pure Vectors

Combining discrete OpenStreetMap polygons with continuous GHSL raster density gives our system multi-layered geospatial context.